VEDANG TRIVEDI

Software Engineer | Data Engineer | Distributed Systems, Python & ML Systems

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Profile Summary

Software Engineer and Data Engineer with expertise in software engineering, distributed systems, SQL-driven query optimization, and applied machine learning for systems efficiency. Designed, built, and shipped two Power Platform applications at Kraft Heinz used daily by **50+ people** across **10 towers**, engineering automation estimated to save **1,600+ hours** a year. Also builds systems-level research: an entropy-aware LLM-routing model that cuts inference cost by **78%**, and a learned cardinality estimator for query-plan optimization.

Technical Skills

Languages: C, C++, Python, SQL, JavaScript
Databases: PostgreSQL, MongoDB, MySQL, Redis
Systems & ML: Distributed Systems, Query Optimization, QLoRA/PEFT Fine-Tuning, XGBoost
App Dev & Automation: Power Apps, Power Automate, Git/GitHub, Figma, LangChain
Data & BI: Power BI, Pandas/NumPy, NLTK/spaCy

Professional Experience

Kraft Heinz — **Feb 2026 – Present**
KHMS/AIT Intern, GCC India — *Ahmedabad, Gujarat*

- Engineered the AIT Power App end-to-end (5 screens, 20 SharePoint columns across 4 data types) after mapping the manual, email/Excel-based opportunity-tracking process, deploying a standardized 7-stage lifecycle model to **50+ users** across **10 towers**.
- Implemented 6+ conditional logic rules (field visibility, record locking, cascading filters, form validation) and 2 automated email-notification triggers, cutting an estimated **1,130+ hours a year** of manual status-update and reporting effort.
- Debugged and resolved **12 documented defects** through root-cause analysis across iterative build-test-fix cycles with real users, shipping the app entirely on existing Microsoft 365 licensing (Power Apps, SharePoint, Power Automate) with zero external budget.
- Migrated Health Check data onto a governed SharePoint backend (**5 lists**) and engineered the Health Check and OpEx Survey app suite – wizard-based flow, dependent dropdowns, submission validation – for **50+ users** across **11 towers**, cutting submission errors to near-zero and saving an estimated **267 hours a year**. Also implemented role-based access control across the platform, adding a limited-permission tier to enforce data governance without admin bottlenecks.

Karma Technolabs — **Jan 2024 – Apr 2024**
Web Development Intern — *Ahmedabad, Gujarat*

- Enhanced search, filter, and upload functionality on a live CRM platform, contributing end-to-end across the development-to-deployment lifecycle.

Projects

UGAD-Lite: Uncertainty-Guided Adaptive Distillation | *ML Systems* — GitHub

- Engineered an entropy-conformal router that dynamically directs queries between a large LLM (GPT-4o) and a fine-tuned Small Language Model (Phi-3-Mini via QLoRA), calibrated with conformal prediction for statistical safety guarantees.
- Cut LLM inference cost and token usage by 78% at a 94.8% task success rate on GSM8K and synthetic arithmetic benchmarks, with a <5% false-negative rate.

CardEst | *SQL, Query Optimization* — GitHub

- Built an XGBoost-based cardinality estimator trained on the TPC-H dataset to improve query-optimizer execution-plan accuracy.
- Cut query latency and resource usage in 70% of evaluated cases by improving Pearson correlation and reducing median estimation error.

Raft Census (DBFS) | *Distributed Systems, Raft Consensus* — GitHub

- Built a language-agnostic, decentralized file-storage system using Raft cluster consensus (Redis + MongoDB-backed), with work-stealing task distribution to support concurrent multi-client create/read/write/delete operations with efficient search and caching.

UML2PlantUML | *LLMs, Machine Learning* — GitHub

- Built a small-scale, local reproduction of the approach from Sadeghi et al. (MODELS 2026, arXiv:2607.28987) for handwritten UML-to-PlantUML generation, preprocessing a dataset of activity and sequence diagram images for model training.
- Fine-tuned and benchmarked 7 vision-language models (ViT-BART, BLIP, Flan-T5, ViT-GPT2, Pix2Struct, CLIP, MiniCPM) to generate PlantUML code from class diagram images, and deployed the trained model on Kaggle, where it has drawn numerous downloads.

Distributed Database System | *2PC, Query Optimization* — GitHub

- Engineered a distributed database with a coordinator-worker topology, implementing horizontal/vertical data fragmentation and a global data dictionary for location-transparent query routing.
- Implemented Two-Phase Commit (2PC) for multi-node transaction atomicity and semi-join query optimization, cutting cross-node bandwidth by 60–70% and query latency by 3–4x versus centralized execution.

Education

DA-IICT (Dhirubhai Ambani Institute of Information and Communication Technology) — **2024 – 2026**
M.Tech, ICT (Software Systems) | CPI: 7.47/10 — *Gandhinagar, Gujarat*

Gujarat Technological University — **2020 – 2024**
B.E., Computer Engineering | CGPA: 7.77/10 — *Ahmedabad, Gujarat*

Positions of Responsibility

Teaching Assistant – Software Engineering & Object-Oriented Programming — **Aug 2024 – Present**
DA-IICT, Gandhinagar — *Gandhinagar, Gujarat*

- Conducted labs and doubt-clearing sessions of 100+ students, designed and evaluated assignments/exams, and mentored students in Software Engineering and Object-Oriented Programming.

Achievements

- GATE CS 2024 qualifier.